

EX. NO: 6 c

DATA VISUALIZATION USING POWER BI (ONLINE RETAIL DATASET)

AIM:

To clean up raw retail store data, bring it into Power BI, and create an interactive dashboard that shows total sales, top products, top countries, and order trends over time.

STEP-BY-STEP PROCEDURE

Step 1: Load the Dataset into Power BI

1. Open **Power BI Desktop** on your computer.
2. On the **Home** ribbon, click on **Get Data** and select **Text/CSV**.
3. Pick your dataset file (OnlineRetail_2.csv or OnlineRetail.csv).
4. Set the File Origin/Encoding to **1252 (ANSI)** or **ISO-8859-1** so that currency symbols and non-English characters display properly.
5. Click **Transform Data** (this opens the **Power Query Editor** screen).

Step 2: Clean and Prepare the Data (Power Query Editor)

Data directly from real-life systems usually has duplicates, blank values, and errors. Clean it using these quick actions:

1. **Remove Duplicate Rows:**
 - Highlight all columns, right-click any column header, and select **Remove Duplicates**.
2. **Remove Cancelled and Bad Transactions:**
 - Click the drop-down arrow next to Quantity → hover over **Number Filters** → select **Greater Than** → type 0.
 - Click the drop-down arrow next to UnitPrice → hover over **Number Filters** → select **Greater Than** → type 0. *(This gets rid of negative quantities, test orders, and cancelled items).*
3. **Fill Missing Information:**
 - Right-click the Description column → select **Replace Values** → replace empty cells with "Unknown Product".
 - Right-click the CustomerID column → select **Replace Values** → replace empty cells with 0.

4. **Set Data Types Correctly:**

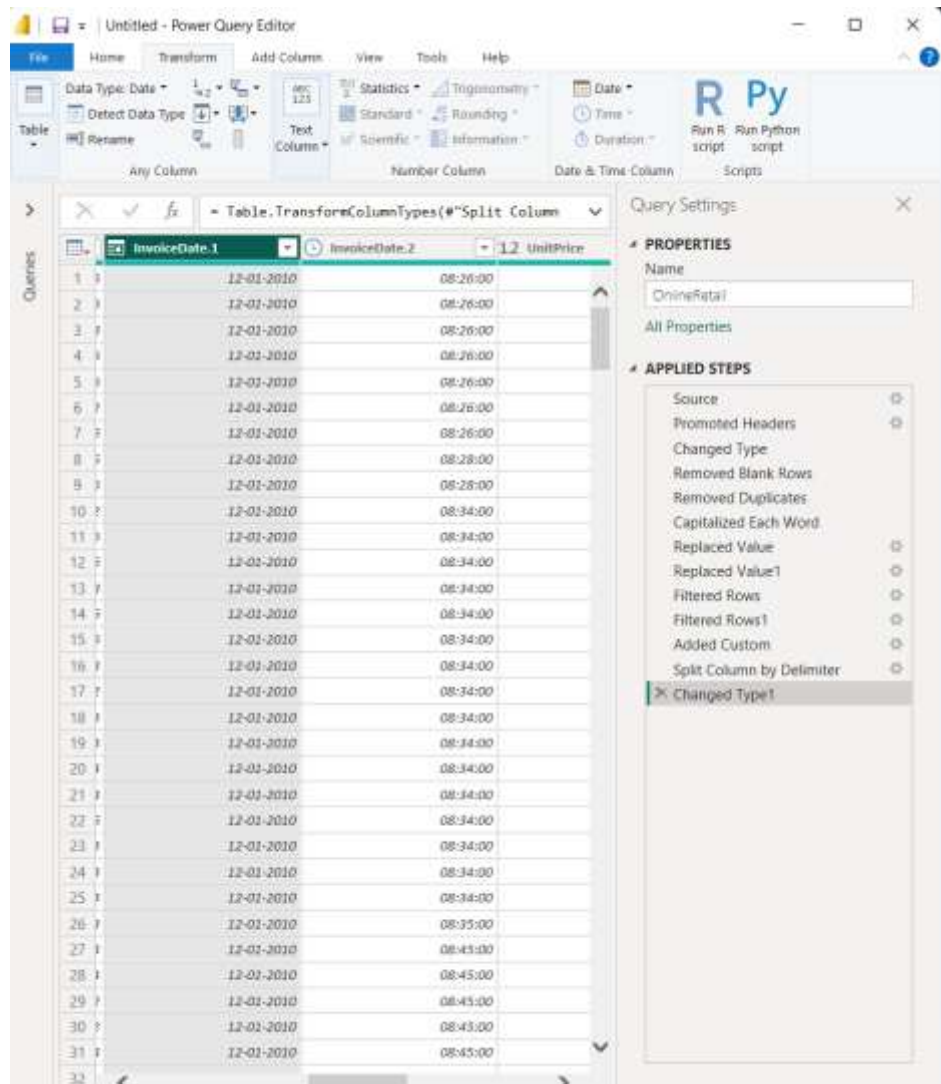
- InvoiceNo, StockCode, Description, Country: Set as **Text**.
- Quantity, CustomerID: Set as **Whole Number**.
- UnitPrice: Set as **Decimal Number**.
- InvoiceDate: Set as **Date/Time**.

5. **Add Total Sales Column:**

- Go to the **Add Column** tab at the top → click **Custom Column**.
- Name it Total Sales.
- Type the formula: [Quantity] * [UnitPrice].
- Click **OK**, then change its data type to **Decimal Number**.

6. **Apply and Load:**

- Click **Close & Apply** on the top left to save changes and load the dataset into Power BI.



Step 3: Create Formulas (DAX Measures)

In the main Power BI window, click on **New Measure** under the **Modeling** tab to build these basic metrics:

- Total Revenue:**

Total Revenue = SUM(OnlineRetail[Total Sales])

- Total Quantity Sold:**

Total Quantity = SUM(OnlineRetail[Quantity])

- Total Number of Orders:**

Total Orders = DISTINCTCOUNT(OnlineRetail[InvoiceNo])

- Average Unit Price:**

Average Unit Price = AVERAGE(OnlineRetail[UnitPrice])

Step 4: Build the Visual Dashboard

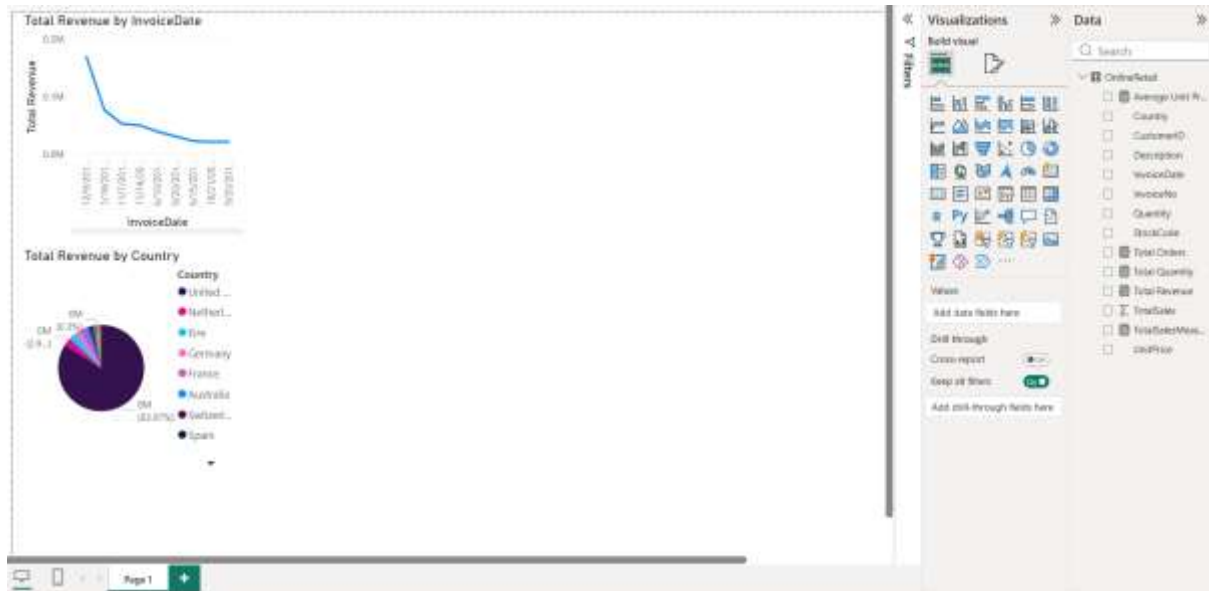
Drag and drop the following visual components from the **Visualizations** panel onto your canvas:

- **1. KPI Summary Cards (Top Section)**
 - Add 4 **Card** visuals across the top row to display immediate summary numbers:
 - Card 1: Total Revenue
 - Card 2: Total Orders
 - Card 3: Total Quantity
 - Card 4: Average Unit Price
- **2. Sales Trend Line Chart (Middle Left)**
 - Add a **Line Chart**.
 - **X-axis:** Drag InvoiceDate (shows sales progression across years/months).
 - **Y-axis:** Drag Total Revenue.



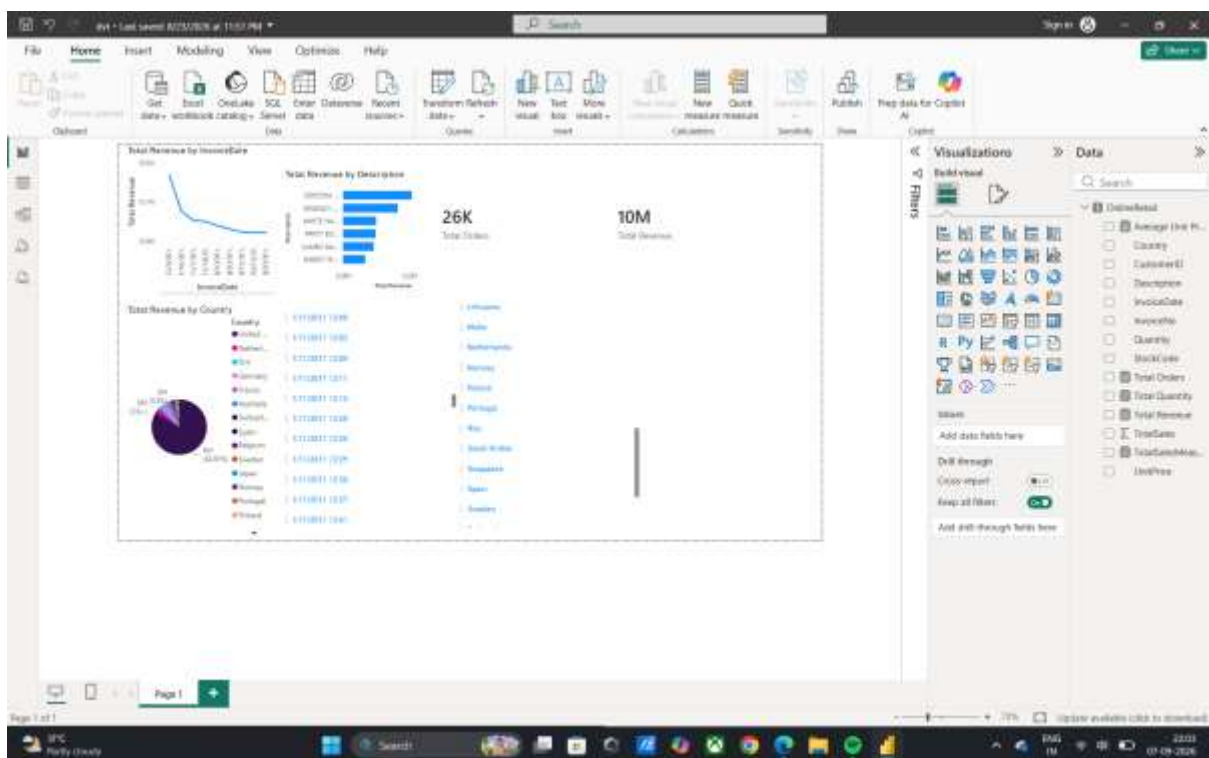
- **3. Regional Sales Breakdown (Middle Right)**

- Add a **Pie Chart** or **Donut Chart**.
- **Legend:** Drag Country.
- **Values:** Drag Total Revenue.



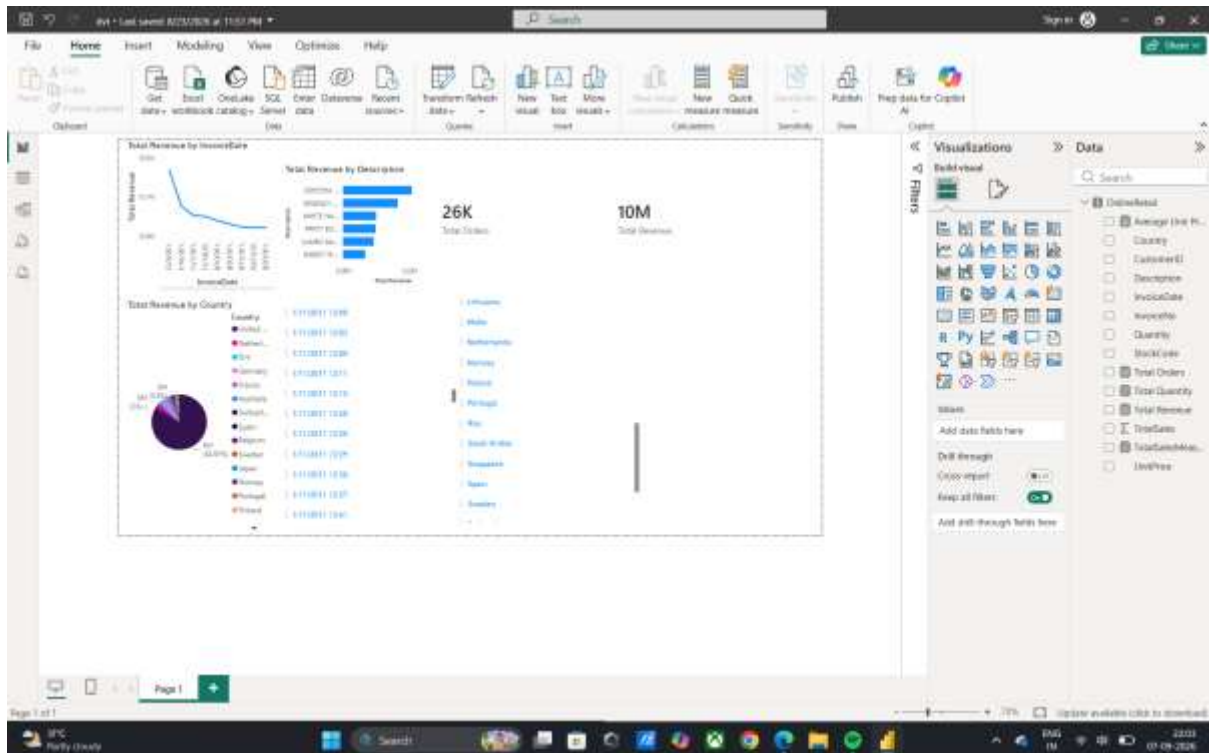
• 4. Top Products Bar Chart (Bottom Section)

- Add a **Clustered Bar Chart**.
- **Y-axis:** Drag Description (Product name).
- **X-axis:** Drag Total Revenue.



- **5. Interactive Slicers (Filters)**

- Add a **Slicer** visual on the side panel and select Country.
- Add a second **Slicer** visual and select InvoiceDate. *(Clicking any country or date range in these slicers will instantly update all charts on your screen!)*



RESULT:

The raw OnlineRetail dataset was cleaned, transformed, and organized into Power BI Desktop. An interactive dashboard was constructed with KPI cards, line charts, bar charts, pie charts, and filter slicers to analyze business performance.